

Alpha Engine

Causal volatility regimes, policy-text sentiment,
and research marks on NIFTY options:
a single-volume technical narrative

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Abstract

This is one document meant to do two jobs at once. It is the technical specification of Alpha Engine—a local research stack that fuses India VIX state with Reserve Bank of India statement language and then paper-trades two-percent out-of-the-money NIFTY options. It is also a self-contained course. A curious reader who has never priced an option should be able to start at Chapter 1 and finish with a working mental model of Garman–Klass variance, the variance risk premium, Ornstein–Uhlenbeck mean reversion, Hamilton Markov regimes, retrieval-augmented local LLMs, Hagan SABR, Kelly sizing, and why a 39 percent win rate can still be a coherent book. Every model is introduced as a sentence a beginner can keep, then derived far enough that a practitioner can argue with the implementation, then placed in a live-market setting where the sentence breaks. The engine is not a broker, not a signal service, and not a claim that India VIX has been forecasted. Historical rupee P&L is a mark-to-model built from VIX and a default SABR smile. That limitation is treated as part of the subject matter, not as a footnote.

Contents

A note to the reader	2
1 What the market is selling you	3
1.1 An index is not a stock	3
1.2 Price, return, and the first lie about risk	3
1.3 Implied versus realized, said slowly	3
1.4 Why India, why text	3
2 A first walk through Alpha Engine	4
3 Garman–Klass and the anatomy of a trading day	4
3.1 Why close-to-close is not enough	4
3.2 The estimator	4
3.3 Why this estimator, not the others	5

3.4	Where GK lies	5
4	The variance risk premium	5
4.1	The sentence to keep	5
4.2	Two definitions, do not mix them	5
4.3	Why a desk cares	6
4.4	Why a desk does not trust it alone	6
4.5	The economic content you should not unlearn	6
5	Mean reversion: the Ornstein–Uhlenbeck view of VIX	6
5.1	The sentence to keep	6
5.2	The stochastic differential equation	6
5.3	How the code estimates (θ, μ, σ)	7
5.4	Why OU rather than a random walk or GARCH	7
5.5	Where OU lies	7
6	Hidden weather: Markov-switching regimes	7
6.1	The sentence to keep	7
6.2	Hamilton’s idea [7], without the textbook fog	8
6.3	Which state is the storm?	8
6.4	Causal refit	8
6.5	Why two states, why not a threshold	8
6.6	The lag you will not code away	8
7	A short-horizon VIX forecast that is allowed to veto	9
7.1	The sentence to keep	9
7.2	The regression	9
7.3	How it accidentally deleted an entire side of the book	9
8	Language as a state: RBI statements and local LLMs	9
8.1	The sentence to keep	9
8.2	Why a number at all	9
8.3	The pipeline, without romance	10
8.4	Why RAG on a small model, why not a 70B API	10
8.5	What sentiment is allowed to do	10
9	Options, smiles, SABR, and the mark that is not a fill	10
9.1	The sentence to keep	10

9.2	Black–Scholes as a translator	11
9.3	Why two percent out of the money	11
9.4	SABR, said as a story	11
9.5	Why this is both a step forward and still a lie	11
9.6	P&L arithmetic	12
10	Sizing, Kelly, and the secondary filter	12
10.1	The sentence to keep	12
10.2	Kelly as a ceiling	12
10.3	Meta-labeling	12
11	The daily decision, written as law	13
12	Map of the repository	13
12.1	Entry points you run	14
12.2	Core modules on the research path	14
12.3	Overlay-only and live-only	16
12.4	Tests	16
12.5	Notebooks	16
12.6	Tables and the processed folder	16
13	Causality, look-ahead, and other ways to fake a Sharpe	17
14	What the empirical blotter actually said	17
15	Real-world rooms this model has not entered	18
15.1	The open	18
15.2	The RBI morning	18
15.3	The crash	18
15.4	The listed weeklies	18
15.5	The other people	18
15.6	The machine	18
16	Why each process was chosen	18
17	A roadmap that is not a wish list	19
18	How a curious reader should practice	19
A	Default parameters	20

B Glossary	20
C References	21

A note to the reader

If you are new to volatility, read straight through and skip nothing until Chapter 6. If you already live on a vol desk, skim Chapters 1–2, then slow down at Chapters 5–8 and 11–13; those are the places where this project made reversible mistakes that look like “more features” when they were really identification errors. If you are here only to operate the repo, Chapter 10 is the map of files; everything else is why those files exist.

I wrote the stack as a research instrument that had to run on a sixteen-gigabyte laptop, ingest RBI PDFs by hand, and refuse to cheat with full-sample Markov states. Those constraints are not embarrassing. They are the reason the document keeps asking *what could have been known at date t ?*

Udayveer Singh Andotra

1 What the market is selling you

1.1 An index is not a stock

NIFTY 50 is a capitalization-weighted basket of fifty large Indian listed companies. When someone says “NIFTY was up one percent,” they mean the basket’s official close rose one percent, not that every name did. Options on NIFTY are cash-settled contracts on that basket. You never take delivery of fifty stocks. You take (or pay) the difference between the strike and the settlement value.

That fact matters more than it sounds. A short put on one company can be a story about accounting fraud. A short put on NIFTY is a story about the Indian economy, global dollar liquidity, foreign-portfolio flows, and whatever the Reserve Bank said last Wednesday. Alpha Engine is built on the second story.

1.2 Price, return, and the first lie about risk

Let S_t be the NIFTY close. A simple return is $R_t = S_t/S_{t-1} - 1$. Beginners are taught that risk is the standard deviation of R_t . That is a starting point, not a destination. Two series can have the same standard deviation and completely different lives: one that dribbles $\pm 0.4\%$ every day, and one that sits still for a month and then gaps 4% on a Sunday night in GIFT Nifty.

Volatility is the *scale* of movement. Direction is a separate question. You can lose money being right on direction if you bought insurance that was too expensive. You can make money being sloppy on direction if you bought insurance that was too cheap and the weather turned. Most of this document is about that second sentence.

1.3 Implied versus realized, said slowly

Stand in the NSE options pit—or, more honestly, in front of a screen that shows the option chain—and pick a one-month at-the-money put. It has a price in rupees. Black–Scholes [6] (Chapter 8) is a translator: it converts that rupee price into a number called implied volatility, quoted in annualized percent. If the translator says 16, the market is charging as if NIFTY were going to move like a 16% annualized process for the next month.

Implied volatility is a *price*. It is not a forecast that the Bureau of Statistics will later print 16%. What actually happens to NIFTY over that month, annualized, is realized volatility. The difference between what you were charged and what occurred is the seed of the variance risk premium (Chapter 4).

India VIX is NSE’s attempt to summarize that charge in one official number [14], built from a strip of out-of-the-money NIFTY options in the spirit of CBOE VIX [13]. When this document says V_t , it means that index, in percent. When the code says `indiavix`, it usually means Yahoo’s `^INDIAVIX`, which is a cousin of the official series—close on quiet days, untrustworthy on gaps. That cousin relationship is a first-class limitation.

1.4 Why India, why text

A US vol book has VIX futures, a deep options tape, and a twenty-year academic literature. An India book has a shorter official VIX, a less liquid wing, and a central bank that still writes long

monetary-policy statements in full sentences. Alpha Engine exists because those sentences are not decoration. “Withdrawal of accommodation” and “support the growth impulse” are different permissions for a short-put book. The engine’s unusual choice is to treat the PDF as a state variable next to V_t , not as a blog post you read after the trade.

2 A first walk through Alpha Engine

Before any formula, here is the job the software actually does on a good day.

You download NIFTY open-high-low-close and India VIX from Yahoo, starting at a date you own (ALPHA_MARKET_START). You compute how turbulent the cash session was (Garman–Klass), how that compares with VIX (VRP), how extended VIX is versus its own mean (Ornstein–Uhlenbeck z-score), and the probability that you are in the jumpy hidden state of a two-regime Markov model. Separately, you drop RBI PDFs into a folder. A small local language model reads retrieved passages and emits a hawkish/dovish score in $[-1, 1]$. That score is stamped with a calendar date and painted forward until the next statement.

Once a day those objects are joined into a single **Decision**: buy a put, buy a call, sell a put, sell a call, or stand aside, plus a conviction in $[0, 1]$ that scales position size. Size is a slice of *current* equity, not a fantasy notionally reset every morning. The option that is imagined is a 30-day European, two percent out of the money, marked with Black–Scholes after a SABR smile has been draped over that day’s VIX. It is held for a handful of sessions or until the decision flips, then closed. Rupees are points times lots times the NIFTY lot size.

If that paragraph felt dense, it is supposed to. The rest of the document unpacks each clause until it is boring.

3 Garman–Klass and the anatomy of a trading day

3.1 Why close-to-close is not enough

The textbook variance of daily returns uses only C_t and C_{t-1} . Consider a session that opens at 22,000, trades as high as 22,450 and as low as 21,720, and closes at 22,020. The close-to-close return is almost nothing. Nobody who sat through that session would call it quiet. Parkinson (1980) noticed that the high–low range is a cleaner thermometer of intra-day diffusion. Garman and Klass [5] added the open and close, building on the high–low estimator of Parkinson [4].

3.2 The estimator

Write $h_t = \ln H_t$, $\ell_t = \ln L_t$, $o_t = \ln O_t$, $c_t = \ln C_t$. A standard GK form is

$$\hat{\sigma}_{\text{GK},t}^2 = \frac{1}{2}(h_t - \ell_t)^2 - (2 \ln 2 - 1)(c_t - o_t)^2. \quad (1)$$

The first term rewards a wide range. The second term subtracts the part of that range that is already explained by a trend from open to close, so a one-way drift is not double-counted as “volatility.”

Alpha Engine averages these daily variance prints over twenty-one sessions, multiplies by 252

to annualize, and takes a square root. That number, σ_t^{rv} , is a decimal (so 0.12 is twelve percent). India VIX is quoted in percent. The code therefore compares V_t with $100 \cdot \sigma_t^{\text{rv}}$, not with σ_t^{rv} itself. Unit mistakes here look like a “broken VRP” and are not philosophical.

Extreme GK values are clipped. A bad tick that prints a low of 1 rupee would otherwise dominate the twenty-one day window and announce a crisis that did not happen.

3.3 Why this estimator, not the others

Parkinson uses only high and low and is statistically efficient under ideal Brownian motion. It is blind to opens and overnight gaps. Yang and Zhang [8] is the grown-up estimator when overnight jumps are first-class, which they increasingly are for NIFTY because GIFT Nifty and SGX trade when Mumbai is closed. Alpha Engine stays on GK because the original mandate was a cash-session thermometer with no extra moving parts. If you rebuild this book for 2026-and-after live trading, Yang–Zhang is the correct next estimator, not a different religion.

3.4 Where GK lies

The derivation assumes continuous diffusion and observed highs and lows. Circuit breakers censor the range: the official high/low stop at the limit, so GK *understates* variance on the days you most care about. Opening auctions, tick size, and a last-price bounce add microstructure noise the other way. On a limit-down day the engine can print a modest realized vol, a fat VRP, and a story that says “implied looks rich” when the truth is “the tape was closed.” A desk would not sell variance that afternoon on a GK reading.

4 The variance risk premium

4.1 The sentence to keep

Implied vol is what insurance costs. Realized vol is what the accident report says. When insurance regularly costs more than the accidents, selling insurance has a positive average—and a left tail that pays for that average.

4.2 Two definitions, do not mix them

The code stores a *volatility-point* gap

$$\text{VRP}_t = V_t - 100 \cdot \sigma_t^{\text{rv}}. \quad (2)$$

If VIX is 16 and twenty-one day realized is 12%, $\text{VRP}_t = 4$. Positive means “implied looks rich versus the recent past.”

The academic variance risk premium [12, 11] is a statement about two probability measures:

$$\text{VRP}_t^{\text{var}} = \mathbb{E}_t^{\mathbb{Q}} \left[\int_t^{t+\tau} \sigma_u^2 du \right] - \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{t+\tau} \sigma_u^2 du \right]. \quad (3)$$

VIX squared is a (noisy, thirty-day) estimate of the first expectation. Subsequent realized variance

is a noisy outcome of the second. They are cousins, not twins. This document uses the code’s definition when discussing the repo and the academic one when discussing literature.

4.3 Why a desk cares

You do not need a stochastic-volatility model to ask “am I being paid to sell this?” VRP is the cheapest answer that uses only VIX and the cash index. It does not need an option chain, which Yahoo will not give you for 2016.

4.4 Why a desk does not trust it alone

Realized vol is backward-looking. After a month of sleep, VRP looks delicious the morning a geopolitical headline arrives. After a crash, realized vol stays high while VIX is already sliding back toward 16, so VRP looks thin precisely when selling vol is starting to work again. Alpha Engine therefore refuses to let the sign of VRP cast a majority vote. In an earlier draft it did (a 0.45 weight). The blotter became a put-buying machine because GK realized often sat above Yahoo VIX. That was not “the market telling us vol is cheap.” That was one feature bullying the others.

4.5 The economic content you should not unlearn

A structurally positive variance risk premium on equity indices is one of the most replicated facts in empirical finance [12, 11]. It is compensation for states of the world in which wealth is already down and you are being asked to pay on short puts. A year with a 25% hit rate and a hole in 2020 is consistent with that story. A year with a 60% hit rate on a three-month window is not evidence you found a better premium; it is evidence you sampled a quiet tape.

5 Mean reversion: the Ornstein–Uhlenbeck view of VIX

5.1 The sentence to keep

VIX is not a stock. It is pulled toward a long-run level. Spikes fade. That pull has a speed and a typical size. The z-score is how many typical sizes you are away from the pull’s target *as estimated using only the past*.

5.2 The stochastic differential equation

$$dV_t = \theta(\mu - V_t) dt + \sigma dW_t. \quad (4)$$

Here μ is the attractor, $\theta > 0$ is the strength of the spring, σ is residual noise, and W_t is Brownian motion. The same linear mean-reverting diffusion appears in Ornstein and Uhlenbeck [1] and, in interest-rate form, in Vasicek [3]. If $V_t > \mu$, the drift is negative: fear is expected to ease. If $V_t < \mu$, the drift is positive: fear is expected to rebuild toward normal.

The half-life of a deviation is $\ln 2/\theta$. One calibration on this project printed $\mu \approx 16.5$ and a half-life of a fraction of a year. In English: India VIX forgets, and it forgets on a horizon of months, not decades.

The stationary variance of an OU process is $\sigma^2/(2\theta)$. That is the natural denominator for a z-score

$$z_t = \frac{V_t - \hat{\mu}_t}{\sigma_t / \sqrt{2\hat{\theta}_t}}. \quad (5)$$

5.3 How the code estimates (θ, μ, σ)

In discrete time a first-order autoregression

$$V_t - V_{t-1} = a + bV_{t-1} + \varepsilon_t \quad (6)$$

maps to θ and μ by matching drift terms. The residual scale maps to σ . This is ordinary least squares, not a drama.

The drama is the window. If you estimate μ on 2015–2026 and then compute z_{2016} , you have told 2016 about COVID. Alpha Engine uses an *expanding* window: at t , only $\{V_s : s \leq t\}$ is visible. The first year of z-scores is noisy because μ is still being learned. That noise is the price of honesty.

5.4 Why OU rather than a random walk or GARCH

A random walk on VIX says last week’s fear is this week’s starting point with no spring. Empirically the spring is the whole point. GARCH models the variance of *NIFTY returns*. That is a different object from the level of VIX. You can have GARCH on returns and OU on VIX in the same book. They are not substitutes.

5.5 Where OU lies

The Gaussian increment is wrong. VIX jumps up faster than it crawls down. An OU z-score treats a two-day spike to 28 as “four standard deviations, so fade it” at the exact moment a trader’s job is not to fade it. That is why the decision layer has a separate *breaking* rule on five-day percentage change and raw VIX level. The z-score is the spring. The breaking rule is the jump.

OU is also slow to admit that μ itself moved. The 2013–2016 India vol regime and the 2021–2024 low-vol regime do not share a soul. An expanding OLS μ is a compromise between “use all history” and “forget 2013.” A rolling ten-year window, or a time-varying μ_t as a slow random walk, is the next honest model.

6 Hidden weather: Markov-switching regimes

6.1 The sentence to keep

Some months are a different planet from other months. A two-state Markov chain is a weather model: today’s state makes tomorrow’s state more likely to be the same, and each state has its own typical VIX level and its own typical gustiness.

6.2 Hamilton’s idea [7], without the textbook fog

Let $R_t \in \{0, 1\}$ be unseen. Conditional on the state,

$$\tilde{V}_t \mid R_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad (7)$$

where $\tilde{V}_t$ is standardized VIX (the engine scales before fitting so numerical optimization does not fight units). The state evolves as

$$\mathbb{P}(R_t = j \mid R_{t-1} = i) = P_{ij}. \quad (8)$$

A large P_{00} means calm weather persists. A small P_{01} means you do not jump to the storm every other Tuesday.

What you can compute at the close of day t , using only $V_1, \dots, V_t$, is the filtered probability

$$p_t = \mathbb{P}(R_t = \text{high-vol} \mid V_1, \dots, V_t). \quad (9)$$

What you must not use in a backtest is the smoothed probability $\mathbb{P}(R_t \mid V_1, \dots, V_T)$, which has read 2020 while labelling 2018.

6.3 Which state is the storm?

The optimizer names states 0 and 1 as it pleases. Early code assumed “state 1 is high vol” and, when parameter names were missing, defaulted to that guess. The current code compares the two residual variances—or, if names are absent, the probability-weighted variance of VIX in each state—and assigns the storm label to the larger one. If you ever see a dashboard where “high vol” lights up in a dead tape, look here first.

6.4 Causal refit

The engine refits roughly every twenty-one sessions on data up to now. That is slower than an ideal sequential Bayesian update and faster than a once-and-forever 2015–2026 fit. Twenty-one is a trading month, not a theorem.

6.5 Why two states, why not a threshold

“High vol if $V_t > 20$ ” treats $V_t = 19$ after a month at 13 as identical to $V_t = 19$ after a month at 28. The hidden state is persistence. Two states are coarse—2018 emerging-market wobble, 2020 COVID, and 2022 global hiking are not the same storm—but three states on three thousand daily points overfit the last crisis and then look brilliant in-sample.

6.6 The lag you will not code away

Filtered probabilities update after the first shock print, not before it. The Monday that VIX goes from 13 to 22 still “looks calm” at Monday’s open in any causal filter. That is why Alpha Engine will not let $p_t \geq 0.5$ be the only permission to buy a put. The breaking rule exists to cover the first day the filter is late.

7 A short-horizon VIX forecast that is allowed to veto

7.1 The sentence to keep

The project does not predict “VIX tomorrow equals 17.4.” It predicts a five-session *percentage change* with a ridge regression trained only on changes that have already realized, and then uses that number as a veto: do not sell vol if the forecast says fear is about to rise; do not buy vol if the forecast says fear is about to fall—unless the tape is already breaking.

7.2 The regression

Features at t are the OU z-score, VRP, the five-day realized VIX change, and p_t . The label is $V_{t+5}/V_t - 1$. At date t the model is fit on rows whose label was known, i.e. whose $t' + 5$ is already in the past. Coefficients are L_2 -shrunk (ridge) because four correlated thermometers of the same object will otherwise fight.

This is a small, causal, linear nowcast. It is not HAR-RV, not a neural net, and not a VIX futures curve. It exists because a rules book that never asks “and then what?” will sell vol into an already-rising tape.

7.3 How it accidentally deleted an entire side of the book

On a multi-year decline in India VIX, a one-year percentile says “cheap” most days, and a ridge trained on mean-reverting data says “a rise is coming” most days. Combined, the forecast veto cancelled almost every intended short. The blotter became long puts. One version of that book made about eight percent with a six percent drawdown and a 39% hit rate—convexity working. Loosening the veto to “force shorts back in” made the same book lose money. The lesson is not “forecasts are bad.” The lesson is that a veto is a position, and on this sample that position *was* the short-vol side.

8 Language as a state: RBI statements and local LLMs

8.1 The sentence to keep

A monetary-policy PDF is a dated event. The engine compresses it into a number $s_t \in [-1, 1]$ and then refuses to pretend that number existed before the date on the file.

8.2 Why a number at all

You cannot put five thousand words into a position sizer. You can put a stance. By convention here, positive is hawkish (inflation, tightening, withdrawal of accommodation) and negative is dovish (growth, easing, accommodation). Zero is not “the repo was unchanged.” Zero is “the language does not lean.” Unchanged repos with hawkish paragraphs should not score zero; an early prompt that used 0.0 as the JSON example taught small models to copy 0.0 and then write a clever sentence. That bug is documented because it will return the moment someone pastes an example score into a prompt again.

8.3 The pipeline, without romance

PDFs live in `data/raw/mpc_archive/`. Automatic download from the RBI site was attempted, then removed, because a laptop job that depends on a brittle scrape will fail silently and you will think the model is broken. You drop files. Filenames should carry YYYY-MM-DD (year-month-day, never year-day-month).

The file is split into overlapping character chunks. Each chunk is embedded with MiniLM, a small sentence transformer that turns a paragraph into a few hundred numbers such that similar paragraphs sit nearby. Those vectors live in Chroma. At score time the engine asks for chunks about repo, inflation, growth, hike, cut, accommodation—not for the title page.

A local Ollama model sees those chunks and must return JSON. The call has a timeout measured in tens of seconds. If Ollama is not running, or if `llama3.1:8b` has eaten thirteen gigabytes of a sixteen-gigabyte machine, the call hangs until the timeout and a keyword tilt (hawkish minus dovish phrase counts) is used so the pipeline does not die.

The score is written to DuckDB with its date. Every later trading day inherits it until a newer dated file arrives. That is how a quarterly statement becomes a daily column without smearing it backward onto 2015.

8.4 Why RAG on a small model, why not a 70B API

Retrieval-augmented generation is a long name for “do not paste the whole PDF.” Small models lose the plot in long contexts. They also physically do not fit next to Chrome and a Python process on a sixteen-gigabyte Mac. The right local models for this job were in the three-to-four-billion parameter class. Quality drops. Completeness of the run rises. For a stance in $[-1, 1]$, that trade is acceptable. For legal analysis of the statement, it is not.

8.5 What sentiment is allowed to do

In the connected decision object, s_t does not open a trade. Vol state opens or refuses the trade. Sentiment picks the wing and, if the statement is less than fifteen sessions old, adds conviction. A dovish score can veto a short put when the tape is already nervous. A hawkish score can steer a rich-vol day toward selling calls rather than puts.

If you have one May 2026 PDF, then $s_t = 0$ for the decade before that date. The “RBI-aware book” you think you are testing is a vol book with a 2026 sticker. Populating the archive is the highest-leverage enhancement in Chapter 13 because it requires no new mathematics.

9 Options, smiles, SABR, and the mark that is not a fill

9.1 The sentence to keep

An option price is a compressed view of a distribution. Black–Scholes assumes that distribution is lognormal with one volatility. The market disagrees, especially on downside strikes. SABR is a way to let volatility change with strike. Alpha Engine uses Black–Scholes as the cash register and SABR as the smile draped over today’s VIX, because history does not contain a free NSE chain.

9.2 Black–Scholes as a translator

With spot S , strike K , rate r , tenor T , volatility σ ,

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (10)$$

$$C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad (11)$$

$$P = Ke^{-rT}\Phi(-d_2) - S\Phi(-d_1). \quad (12)$$

Φ is the standard normal distribution function. Put-call parity

$$C - P = S - Ke^{-rT} \quad (13)$$

is a check, not a decoration. The unit tests of this repo include that check.

Tenor in the blotter starts at thirty calendar days and bleeds while the trade is held. Rates default to a constant 6.5% rupee cash rate. That is a research convenience, not the overnight MIBOR path.

9.3 Why two percent out of the money

Two percent is close enough to feel the spot and far enough to be a wing. Strikes are rounded to fifty NIFTY points because listed strikes live on a grid. The lot size defaults to sixty-five, which is a configurable snapshot of the contract unit and will go stale when NSE changes it.

9.4 SABR, said as a story

Let F be the forward. SABR says the forward is driven by a volatility α_t that is itself random, and the two Browns are correlated:

$$dF = \alpha F^\beta dW^{(1)}, \quad d\alpha = \nu\alpha dW^{(2)}, \quad d\langle W^{(1)}, W^{(2)} \rangle = \rho dt. \quad (14)$$

$\beta = 1$ is the lognormal backbone used here (equity index default). $\rho < 0$ is the leverage effect: spot down, vol up, which is why OTM puts trade rich. ν is how violently the smile flexes.

Hagan’s expansion [9] gives an implied vol $\sigma_{\text{SABR}}(K)$ in closed form. For a broader reading of the surface see Gatheral [10]. Alpha Engine sets α from India VIX (the ATM height), freezes $\rho \approx -0.35$ and $\nu \approx 0.75$, and reads σ_{SABR} at the two-percent strike. Then Black–Scholes is called with that σ , not with raw VIX.

9.5 Why this is both a step forward and still a lie

Flat Black–Scholes on VIX prices a two-percent put as if it had ATM vol. Real OTM puts are richer. In a crash the paper mark of a long put is therefore *too small*, and the paper mark of a short put is *too kind*. Both biases flatter the wrong side of a disaster. A live calibrator to today’s NSE chain—which the repo also contains—is the correct tool for *today*. It cannot mark 2017 unless you stored 2017’s chain.

Sprint 1 of this project shipped SABR and a live fetcher. That made it feel as if SABR were

“already running” on the historical blotter. It was not. Two pipes existed. Only later were they joined, and even then only as VIX-height plus default shape.

9.6 P&L arithmetic

A long option:

$$\Pi = (P_{\text{exit}} - P_{\text{entry}} - 2c) \times \text{lots} \times 65, \quad (15)$$

with $c = 0.5$ points a side as a toy spread. Shorts flip the difference. Equity updates. The next trade sizes off the new equity. If equity falls through thirty-five percent of start, the engine stops. This is not SPAN margin. It is a fuse so a short-put fantasy cannot print crores of losses on a ten-lakh notionally reset book—which an earlier version did.

10 Sizing, Kelly, and the secondary filter

10.1 The sentence to keep

Bet a small, known fraction of the pile you actually have. Do not bet a fraction of a pile you had in 2015. After you have lost enough times in the same weather, consider skipping.

10.2 Kelly as a ceiling

For a binary bet with win probability p and payoff ratio b (win size over loss size),

$$f^* = p - \frac{1-p}{b}. \quad (16)$$

Kelly [2] derived f^* as the growth-optimal fraction. Full Kelly maximizes asymptotic growth and also maximizes the chance you look like an idiot in any finite sample. Half-Kelly is the industry’s shrug. Alpha Engine computes a Kelly-like fraction and then mostly ignores it in favour of a 1.2% equity cap, because estimated p on two-percent NIFTY wings is not a number you should lever.

Short options use three times premium as the unit of risk. That is a haircut, not a model of SPAN. A real short-put book in March 2020 was a margin event, not a premium event.

10.3 Meta-labeling

López de Prado’s [15] useful idea is to split “should a trade exist?” from “should I take this one?” The primary rule proposes. The secondary model, an XGBoost classifier, estimates $\mathbb{P}(\text{this trade wins} \mid \text{features at entry})$. After forty closed trades the engine begins to skip when that probability is below 0.48, and otherwise scales conviction.

Forty observations is a toy. Feature names overlap the primary rule, so the classifier often relearns the rule and then refuses trades the rule already liked. Treat the wiring as a completed homework, not as a demonstrated edge. Meta-labeling starts to matter when the primary book has thousands of bets and a clean label. A six-day two-percent wing marked on VIX is a muddy label.

11 The daily decision, written as law

The function `decide(row)` is the constitution. Models may estimate. This function acts.

Inputs. V_t , its one-year percentile q_t , its five-day change, OU z_t , VRP $_t$, Markov p_t , sentiment s_t , days since the last dated PDF, five-day forecast $\hat{r}_{t,t+5}$.

Vol weather. If V_t is in a shock—at least 24 and still rising, or a violent five-day jump from already-elevated levels—the weather is *breaking*: buy puts. Else if q_t is in the top third of the last year, weather is *rich*: sell premium, unless the forecast veto fires. Else if q_t is in the bottom third, weather is *cheap*: buy premium, unless the forecast veto fires. Else stand aside.

Wing. Hawkish s_t steers shorts toward calls (upside capped by tighter policy) and adds weight to puts when already buying vol. Dovish s_t steers cheap-vol buys toward calls if the tape is calm, and forbids short puts if the tape is nervous.

Freshness. A statement younger than fifteen sessions may raise conviction. It may not invent a trade out of dead vol unless the score is extreme.

Forecast veto. A predicted five-day VIX rise above two and a half percent cancels a short. A predicted drop of the same size cancels a non-shock long. Loosening that veto, on this sample, made the book worse.

What this law is not. It is not an optimizer. It is not a neural net. It is an inspectable sentence so that when the blotter prints one hundred and seventy-six buy-puts and zero sells, you can find the clause that did it (percentile plus veto on a falling VIX decade) instead of adding another library.

12 Map of the repository

This chapter is an inventory, not a greatest-hits list. Every Python module in the tree is named below, with whether the main pipeline imports it. Package `__init__.py` files exist under `src/`, `data_pipeline/`, `features/`, `models/`, `execution/`, and `dashboard/`; they make imports work and contain no strategy.

12.1 Entry points you run

Path	Pipeline job
<code>scripts/run_daily_market_update.py</code>	Yahoo OHLC + VIX $\rightarrow$ GK, VRP, OU z , fuse, Markov $\rightarrow$ DuckDB + data/processed
<code>scripts/run_batch_nlp_pipeline.py</code>	Archive PDFs $\rightarrow$ chunks, embeddings, Olama score, paint s_t , refit regimes
<code>scripts/run_backtest.py</code>	Options blotter via <code>options_pnl</code> ; also runs the index overlay
<code>src/dashboard/app.py</code>	Streamlit terminal on <code>regime_predictions</code> and <code>option_trades</code>
<code>src/visualization/regime_plotter.py</code>	Standalone Matplotlib of VIX vs p_t vs s_t
<code>scripts/run_async_nlp_pipeline.py</code>	Optional: copy one PDF into the archive, then call batch NLP
<code>scripts/download_rbi_pdfs.py</code>	Optional: prints the manual archive path; does not download

12.2 Core modules on the research path

These are imported by daily update, batch NLP, the options blotter, or the dashboard.

Path	Purpose in the pipeline
src/bootstrap.py	Puts <code>src/</code> on <code>sys.path</code>
src/config.py	Paths, tickers, start date, lot, risk cap, Ollama model, SABR defaults
src/data_pipeline/db_utils.py	DuckDB create / overlap-safe insert / load
src/data_pipeline/market_fetcher.py	<code>^NSEI</code> and <code>^INDIAVIX</code> with ticker fall-backs
src/data_pipeline/dates.py	YYYY-MM-DD from filenames and statement text
src/data_pipeline/nlp_scraper.py	Load PDF, chunk, attach statement date
src/data_pipeline/exports.py	Snapshot tables into <code>data/processed/*.parquet</code>
src/features/vol_math.py	Garman–Klass variance and expanding OU z-score
src/features/quantitative_metrics.py	Writes <code>market_features</code> : RV, VRP, z_t
src/features/signal_aggregator.py	Dated <code>sentiment_events</code> ; forward-fill onto the panel
src/features/text_embeddings.py	MiniLM embeddings into Chroma
src/features/vix_forecast.py	Causal 5-day Ridge on VIX returns
src/models/regime_classifier.py	Two-state Markov, causal refit, writes p_t
src/models/regime_utils.py	Identifies which state has larger variance
src/models/llm_sentiment.py	Ollama JSON score, timeout, keyword fall-back
src/models/vol_surface.py	SABR Hagan; <code>sabr_iv_from_vix</code> for blotter marks
src/models/meta_labeler.py	Triple barrier helper + XGB $P(\text{win})$
src/execution/strategy.py	<code>decide()</code> / <code>route_row()</code> : the daily law
src/execution/options_pnl.py	Walk-forward options ledger and equity cap
src/execution/black_scholes.py	European price in index points
src/execution/position_sizer.py	Half-Kelly ceiling (overlay and tests)
src/dashboard/components.py	Plotly VIX / high-vol window / stance chart

12.3 Overlay-only and live-only

Path	Status
src/execution/backtester.py	NIFTY close-to-close overlay used by <code>run_backtest.py</code> . Not option P&L. Ignore its tearsheet when judging the vol book.
src/execution/options_engine.py	Live NSE chain fetch + old <code>route_signal</code> . <i>Not imported</i> by daily update, NLP, options blotter, or dashboard. Optional <code>__main__</code> only.

12.4 Tests

Path	What it locks
tests/conftest.py	Adds <code>src/</code> to <code>sys.path</code> for pytest
tests/test_data_pipeline.py	GK sign, OU recovery of μ , PDF dates, LLM JSON fences
tests/test_models.py	Kelly bounds, Markov label, SABR ATM, triple barrier, BS parity, one connected <code>decide()</code> case
tests/test_strategy.py	Percentile buckets, forecast veto, hawkish call wing, conviction
tests/test_signal_aggregator_api.py	<code>statement_date</code> / <code>override_date</code> alias still exists

Run `PYTHONPATH=src python -m pytest -q`. The aligned suite is twenty-five tests.

12.5 Notebooks

Path	Reads
notebooks/01_market_and_regimes.ipynb	VIX, p_t , VRP, z_t from processed parquet or DuckDB
notebooks/02_rbi_sentiment.ipynb	Archive listing and forward-filled s_t
notebooks/03_options_pnl.ipynb	Blotter by year/action and last <code>Decision</code>

12.6 Tables and the processed folder

DuckDB lives at `data/duckdb/alpha_engine_data.db`:

`historical_market_data`, `market_features`, `sentiment_events`, `fused_features`, `regime_predictions`, `option_trades`, `backtest_results`.

The last table is the index overlay. Do not quote it as the options book.

`data/processed/` is no longer an empty directory. After each main script, `exports.py` writes the tables that exist as parquet (csv if parquet fails). Notebooks prefer those snapshots, then fall

back to DuckDB.

Run order remains: daily market update, then batch NLP, then backtest, then dashboard or notebooks. Sentiment is not computed inside the market job; “zero sentiment events” after a daily update is expected until batch NLP has run.

13 Causality, look-ahead, and other ways to fake a Sharpe

A number is causal at t if a person locked in a room with only information dated $\leq t$ could have produced it.

- Using today’s close to trade today’s close is an end-of-day convention, not high-frequency fiction.
- Expanding OU and trailing Markov are causal. Full-sample smoothed states are not.
- A ridge that trains on V_{t+5} as of date t is not causal. Training on labels that already settled is.
- An LLM score with a filename date of 2026-05-08 applied to 2026-05-07 is a leak. Applied from 2026-05-08 forward is not.
- SABR ρ frozen at -0.35 is not a leak. It is an assumption. Those are different sins.
- Painting one sentiment number across the last thirty days because the model forgot to return a date—an early version—is theatre. It makes a chart. It is not a time series.

The index overlay’s first Sharpe used CAGR in the numerator. That is not look-ahead. It is simply the wrong statistic. It is mentioned here so you do not trust old screenshots.

14 What the empirical blotter actually said

This chapter is the opposite of a marketing page.

When the router defaulted to selling puts whenever p_t was below one half, yearly P&L was negative almost every calendar year from 2015 through 2025. Win rates near fifty percent coexisted with large rupee losses: small collected premia, fat left tails. 2020 was the confession.

When the router was inverted into “buy puts whenever anything looks slightly cheap,” win rates fell to the low thirties and the book still lost money, because the puts that were not in a genuine shock died of theta.

When the law in Chapter 9 was combined with a tight forecast veto, the blotter printed on the order of plus eight percent total return, a profit factor a little above one, a drawdown near six percent, and a hit rate near thirty-nine percent—almost entirely long puts. A few sessions in 2026 paid for a cemetery of expired wings. That is a coherent convex book. It is not a carry book.

Loosening the veto to drag shorts back in moved the same code to a loss and a double-digit drawdown without producing a healthy two-sided mix. Forcing a product the sample does not want is not “completing the strategy.”

The index overlay, throughout, printed a high-forties hit rate, a mid-sixes negative CAGR, and a fifty-percent drawdown. It is a different mandate. Stop reading it.

15 Real-world rooms this model has not entered

15.1 The open

NIFTY's cash open is an auction. India VIX Yahoo bars are daily. The engine is an end-of-day creature. It cannot speak about the first fifteen minutes, which is where a lot of event-vol is born and dies.

15.2 The RBI morning

Vol is usually bid going into a policy announcement. After a hold-and-blah statement it is offered. Scoring the PDF at dinner is a research act. It is not capturing the crush. If you want that trade, you need timestamps, a chain, and permission to be flattened at $T - 1$ whether or not the LLM has finished its sentence.

15.3 The crash

Bids vanish on the wing. The 0.5 point cost is comedy. SPAN margin on short puts becomes the whole job. Long OTM puts pay more than a VIX-height SABR mark will admit. The engine's long-vol drawdowns are debit drawdowns. They are not a simulation of being short.

15.4 The listed weeklies

The living product is a weekly expiry, not a thirty-day Platonic European. Rolling weeklies has a calendar-spread personality the blotter never sees. A rebuild that does not face listed expiries is still a teaching blotter.

15.5 The other people

Vol-target funds, systematic put-write ETFs (in markets that have them), and every CTA that buys weakness in VIX are your classmates. When the bottom tercile of VIX is a crowded long-put, VRP is not a premium; it is a fee you pay each other.

15.6 The machine

llama3.1:8b on a sixteen-gigabyte laptop is not a modelling choice. It is a denial-of-service against yourself. The document records that because "just use a better model" is how jobs hang at midnight.

16 Why each process was chosen

GK exists because the project had OHLC and needed a session thermometer. VRP exists because a desk's first question is payment, not the level of VIX. OU exists because VIX is a sprung process and a z-score is the least dramatic way to say "extended." Markov exists because persistence is not a threshold. Ridge exists because a rules book that never asks "what next?" will sell into a wave. MiniLM and Chroma exist because a four-billion-parameter model cannot read a whole PDF.

Ollama exists because the work had to stay on the machine. Rules exist because a fitted classifier from four thermometers to “buy put” will memorize 2020. SABR-from-VIX exists because a smile was required and a historical chain was not. The equity cap exists because the first blotter sized every trade off a reset ten lakh and invented crores.

Rejected on purpose: full-sample smoothed Markov, Google-dork downloads as a required step, painting a score on the last thirty days, eight-billion-parameter local models as a default, live NSE marks inside a 2015–2024 loop.

17 A roadmap that is not a wish list

Do these in order. The first item beats any new model.

1. Put dated MPC PDFs for 2015–2025 in the archive and rerun batch NLP. Until then the RBI chapter is a 2026 appendix.
2. Replace Yahoo VIX with NSE’s official series. Replace inner-join gaps with an explicit missingness flag, not silence.
3. Store daily option chains if you want SABR to be an estimate rather than a costume.
4. Add an event calendar (MPC, Budget, CPI) and flatten shorts in a window around them whether or not q_t is rich.
5. Rebuild P&L on listed weeklies with bid/ask. Retire the thirty-day imaginary contract as a teaching device.
6. Give VIX jumps their own process (jump-OU or a Hawkes intensity). Stop asking Gaussian OU to fade a spike that is still printing.
7. Only then spend time on three-state Markov, HAR-X forecasts, or a larger meta-labeler.
8. Delete or quarantine the index overlay so two tearsheets cannot be mistaken for one book.

18 How a curious reader should practice

Do not start by editing `decide()`. Start by breaking the thermometers.

Compute GK by hand for one noisy day and for one limit-down day. Fit OU on 2015–2019 and stare at z_{2020} versus the expanding z_{2020} . Flip Markov state labels and watch every hedge become a short. Score one PDF with two local models and write down Δs_t next to the 0.15 hawkish threshold. Turn the forecast veto off, record profit factor and drawdown, and walk away without tuning. Read Hagan on SABR until the $\rho < 0$ sentence is obvious. Read Hamilton until filtered versus smoothed is obvious. Read López de Prado until you are embarrassed by forty training rows.

When those exercises are boring, the repo will look small—which is the point of a one-document course. The remaining difficulty is not Python. It is identification: when is India variance cheap *conditional on* policy path, global vol, and the calendar, in a sample that includes fills? That question is not answered here. It is the question this stack was built to make askable.

A Default parameters

Item	Default
Sample start	2015-01-01
GK window	21 sessions
OU minimum history	252 sessions
Markov states / refit cadence	2 / 21 sessions
Imagined tenor / hold	30 days / 6 sessions
Moneyness / grid	2% OTM / 50 points
Lot	65
Cash rate in BS	6.5%
Risk cap / halt	1.2% of live equity / 35% of start
Wing cost	0.5 points a side
SABR (β, ρ, ν)	$(1, -0.35, 0.75)$
Forecast horizon / veto	5 sessions / $ \hat{r} > 2.5\%$
Meta-label skip	$\hat{p} < 0.48$ after 40 closes
Embedding	MiniLM-L6-v2
LLM	ALPHA_OLLAMA_MODEL in .env

B Glossary

ATM

At the money: strike near spot.

Causal

Computable from information dated on or before t .

Conviction

Number in $[0, 1]$ that scales the risk cap.

Filtered probability

Markov state belief using data up to t only.

Forward-fill

Repeat the last dated sentiment until a new statement.

Half-Kelly

Half the growth-optimal fraction; a humility device.

Hawkish / dovish

Policy language leaning tight / easy.

Implied volatility

The σ that makes Black–Scholes match a market price.

India VIX

NSE’s 30-day implied vol index on NIFTY.

Look-ahead

Using information from after t to decide at t .

Mark-to-model

A price from a formula, not from a bid you can hit.

OTM

Out of the money.

Smoothed probability

Markov belief that has seen the whole sample.

SPAN

Exchange margin algorithm; not “three times premium.”

Theta

The bleed of option value as calendar time passes.

Variance risk premium

Extra charge in implied variance versus realized, on average.

C References

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